

Interactive Visual Analysis of Financial Fraud Patterns

Problem Statement

Financial fraud is a rapidly evolving issue that costs businesses billions of dollars annually. While computational algorithms and machine learning models are frequently deployed to detect anomalies, these systems often lack the transparency needed for human analysts to make confident decisions. This aligns with the core concept that computers working alone are insufficient for modern data challenges, making human in the loop systems necessary. Analysts need to quickly understand the context behind an anomaly to separate actual fraud from legitimate but unusual customer behavior. The specific problem I am addressing is the gap between algorithmic anomaly detection and human sensemaking in financial transaction data.

Proposed Methodology and Information Provision

To address this problem, I will develop an interactive visual interface designed to facilitate analytical reasoning. The project will focus on a business dataset containing financial transactions to model normal and abnormal purchasing behaviors.

First, I will use computational methods to process the data, perform necessary data transformations, and generate risk scores for individual transactions. Next, I will design a visual dashboard that allows users to filter these transactions by risk score, geographic location, and temporal patterns. By synthesizing computational outputs with visual representations, the dashboard will empower analysts to drill down into specific clusters of suspicious activity. This approach will provide users with the contextual information they need to visually identify trends that might indicate coordinated fraudulent attacks, while also helping them easily spot false positives generated by the algorithm.

Tools and Implementation

To emulate a complete research project from implementation to presentation, I plan to use a combination of Python and Power BI. Python will be utilized for the heavy data wrangling and to run preliminary clustering algorithms to flag outliers. The resulting processed data will then be imported into Power BI to construct the interactive visual interface. Building the interactive dashboard in Power BI will also prepare me well for the final project deliverable, which involves creating a VisUMD Medium post based on an independent analysis conducted with this specific tool.